No: 11

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ROUTING AT NETWORK LAYER

0) Simulates static Routing Protocol configurations using Cisco Packet Tracer

AIM:

To configure static routing between different networks using Cisco Packet Tracer and verify communication between the network.

STEPS:

1. Open Cisco packet Tracer
2. Place 2 routers, 2 switches and PC's
3. Connect PCs to switches and switches to routers using cables.
4. Assign IP addresses to all PCs and router interfaces
5. Configure the router interface and enable the using as shutdown
6. Configure static routers using the command
7. Configure static raters on both ~~state~~ routers
8. Set the appropriate default gateway on each PC
9. Use the Ping command to test communication between different network.
10. Use simulation mode to observe the packet path.

RESULT:

That the static routing was successfully configured between the network using Cisco Packet tracer and communication between the networks was successfully verified using the ping command

OUTPUT:

Router 0

Router # show ip route static

S 30.0.0.0/8 [10/10] via 20.0.0.2

S 30.0.0.0/32 [10/10] via 40.0.0.2

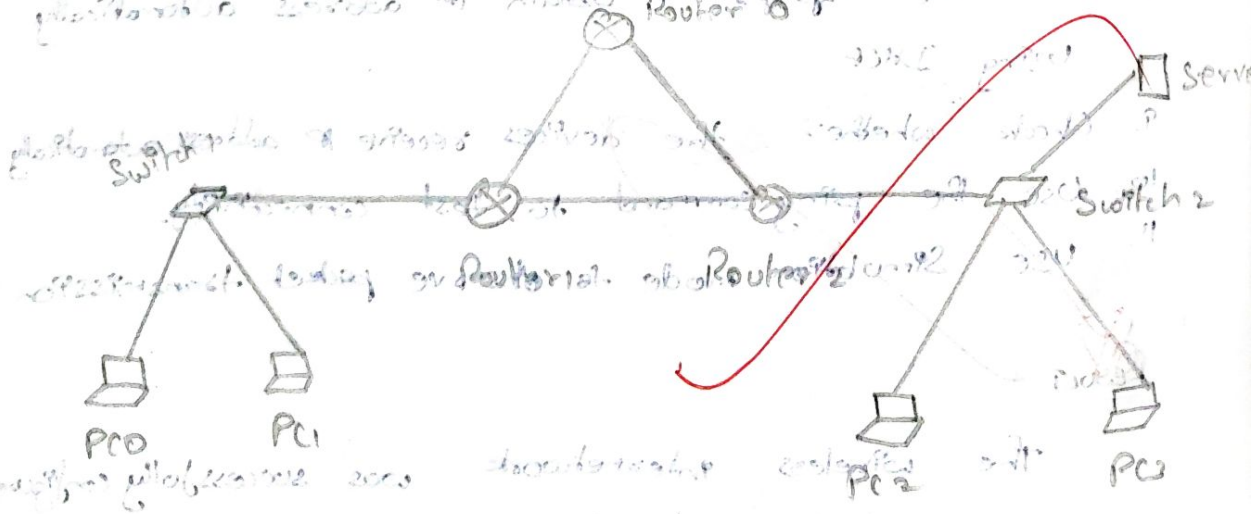
S 50.0.0.0/8 [10/10] via 40.0.0.2

Router 1

Router # show ip route static

S 10.0.0.0/8 [10/10] via 20.0.0.1

S 40.0.0.0/8 [10/10] via 20.0.0.1



b) Simulate RIP using Cisco Packet Tracer

AIM:

To configure and simulate the routing information Protocol (RIP) between routers using Cisco Packet tracer and verify network connectivity.

STEPS:

1. Open Cisco packet tracer
2. Place 2 or 3 routers, switches and PCs
3. Connect the devices using Ethernet cables.
4. Assign IP addresses to PC's and router interfaces.
5. Enable all router interface using no shutdown
6. Configure RIP on the routers
7. Configure the appropriate network addresses on each router according to the topology.
8. Check the routing table using: show ip route
9. Use the ping command to test communication between PC's in different networks
10. Use Simulation mode to observe how RIP learns and select routers.

RESULT:

RIP was successfully configured between the routers using cisco packet tracer. The routes learned the required network routers dynamically and communication between different networks was successfully verified

ping 10.0.0.2

pinging 10.0.0.2 with 32 bytes of data

Request timed out

Reply from 10.0.0.2 bytes=32 time = 3ms TTL=128

Reply from 10.0.0.2 bytes = 32 time = 3ms TTL=128

Reply from 10.00.2 bytes = 32 time = 3ms TTL = 128

ping statistic for 10.0.0.2

packet : sent = 4, Received = 3, lost = 1

Approximate round trip time is milli-seconds

Minimum = 3ms, Maximum = 3ms, Average = 3ms